

Yelp Dataset Sentiment Analysis

Literature Review Survey

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Background and Motivation

- Online platforms like Yelp contain millions of reviews, but the link between text and star ratings is not straightforward.
- Automating star prediction helps users and businesses understand sentiment quickly, and researchers analyze which aspects most affect ratings.
- Literature shows high accuracy for binary sentiment, but more nuanced star prediction is difficult due to imbalanced data, reviewer subjectivity, and noisy language.

Restaurant: XYZ, Kitchener, N2G4Z6, Canada

Rating: ● ● ● ○ ○

I've been to XYZ a bunch of times. It's a decent place. Nice food, lots of variety! The place is really small though, so you almost never find a spot to sit and eat. The service is also slow at times.

Source: Yelp Dataset Challenge: Review Rating Prediction, Nabiha Asghar

Research Directions in Literature

1. **Binary Sentiment Classification** – positive vs. negative reviews.
2. **Star Rating Prediction** – predict 1–5 stars from text.
3. **Bias Analysis** – explore inconsistencies between text and rating.
4. **Aspect-based sentiment analysis (ABSA)** — how does the reviewer feel about certain “aspects” of the product/restaurant (e.g. good food, bad service). Can we use these extracted opinions in our model?

Objectives

- Accurately predict a restaurant's star rating (1–5) from **review text alone**.
- **Compare multiple feature extraction methods** (n-grams, TF-IDF, sentiment lexicons, embeddings).
- Explore **supplemental features** (review length, punctuation usage) and **address class imbalance**
- Benchmark baseline ML models against deep neural networks and ensemble approaches.

Dataset

- Yelp Dataset Challenge (~7–8 million reviews).
- Each review includes text and star rating (+ many more features); known imbalance with most ratings ≥ 4 stars.
- Similar datasets in research: Amazon, Twitter, IMDB for method comparison.
 - Some papers used these

Preprocessing Methods

- Lowercase conversion, punctuation/special character removal, stop word elimination.
- Extract features (length, punctuation) before cleaning.
- Negation handling (“not good” → “not_good”), optional stemming.
- Part of Speech tagging

Feature Engineering Approaches

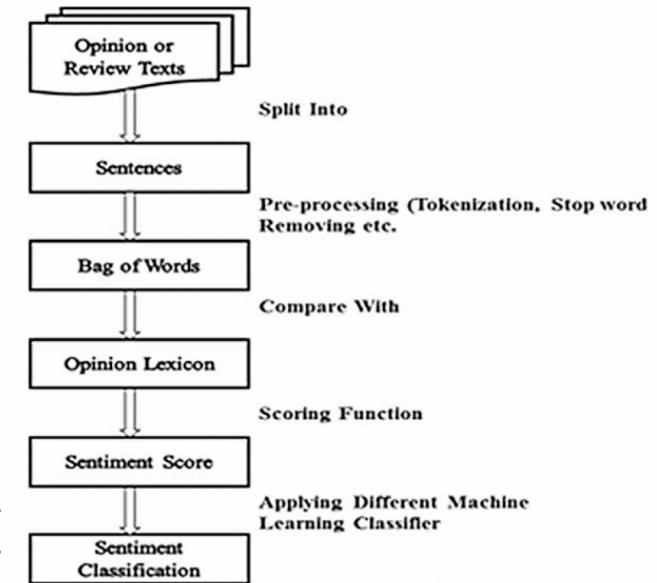
- **Bag of Words (BoW):**
 - Baseline—simple but effective
- **TF-IDF** and **n-grams** (unigrams, bigrams, trigrams):
 - Define rare/meaningful words/phrases
- **Sentiment Lexicons:**
 - Compare text with predefined word lists (e.g., Bing Liu)
 - Struggle with slang/typos
- **Part-of-Speech Tagging:**
 - e.g. use only adjectives
 - improves interpretability and model performance
- **Contextual Embeddings:**
 - e.g. BERT
- Try adding extracted features like review length, positivity/negativity scores, user votes if feasible

Modeling Approaches

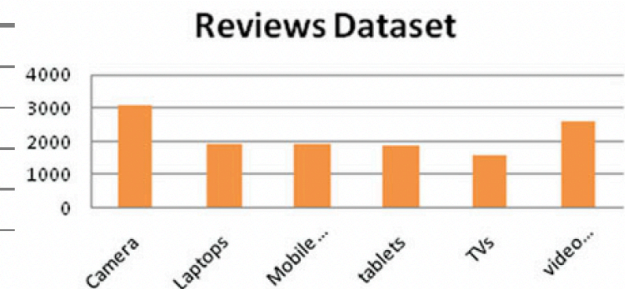
- **Baseline classifiers:** Logistic Regression, Naive Bayes, SVM (linear/non-linear), Passive-Aggressive (similar to SVM).
- **Tree-based ensembles:** Random Forest, AdaBoost.
- Shallow/deep **Multi-Layer Perceptrons** with contextual embeddings.
- **Ensemble methods:** weighted voting, stacking.
- **Ordinal models** (ordered logistic regression) to honor rating order (i.e. 1 is closer to 2 than 4).
- **Regressors** (RandomForest, SVM, etc.), layered strategies (classify sentiment then regress rating).

Sentiment Analysis on Product Reviews Using Machine Learning Techniques; Jagdale et al. (2019)

- **Amazon product reviews:** Camera, Laptops, Mobile phones, tablets, TVs, video surveillance (approx 13k total reviews)
- **Binary** classification
- **Future work:** aspect level analysis could improve (e.g. Camera's quality, megapixel, picture size, structure, lens, picture quality, etc.)



Dataset	Classifiers					
	Naïve Bayes			SVM		
	Accuracy	Precision	F score	Accuracy	Precision	F score
Camera	98.17	98.30	99.03	93.54	93.58	96.66
Laptops	90.22	90.01	94.74	88.16	88.52	93.71
Mobile phones	92.85	91.64	95.64	92.85	91.64	95.64
Tablets	97.17	98.73	98.31	84.12	84.31	91.37
TVs	90.16	90.17	94.72	88.49	85.56	93.89
Video surveillance	91.13	89.95	94.71	79.43	84.25	88.53

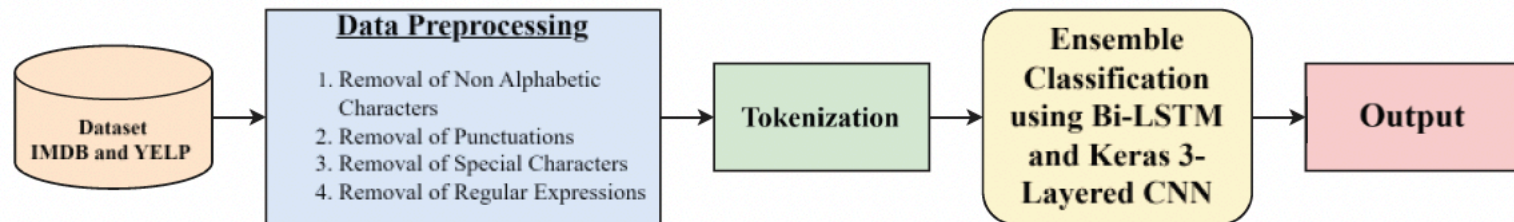


Ensemble Sentiment Analysis Using Bi-LSTM and CNN;

Puneet Singh Lamba et al. (2024)

- **Dataset:** IMDB movie reviews and Yelp (separate models)
- **Binary** classification
- **Tokenization:** NLTK library's word_tokenize()
- **Hybrid** of two techniques (Keras Sequential CNN and Bi-LSTM):
 - Built/saved model with **Bi-LSTM**
 - Built/saved **CNN** with Bert Tokenizer and Keras Module
 - **Ensemble** of both models
 - Weighted average
 - Stacking

Techniques	F1-score	Positive Class F1-score	Negative Class F1-score	Roc_auc Score
IMDB				
Bi-LSTM	0.89	0.89	0.89	0.88
Keras Sequential CNN	0.89	0.90	0.89	0.89
Weighted Average Ensemble	0.92	0.93	0.92	0.96
Stack Ensemble	0.90	0.91	0.92	0.92
YELP				
Bi-LSTM	0.81	0.86	0.71	0.80
Keras Sequential CNN	0.86	0.90	0.77	0.83
Weighted Average Ensemble	0.89	0.92	0.79	0.91
Stack Ensemble	0.90	0.91	0.78	0.90

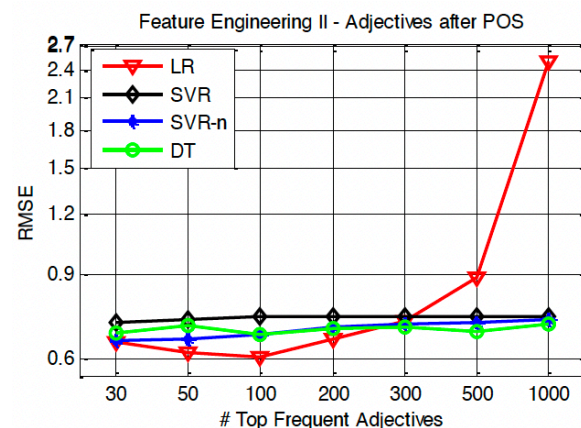
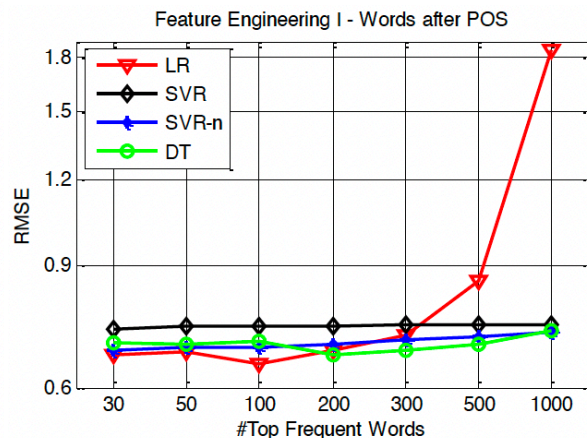
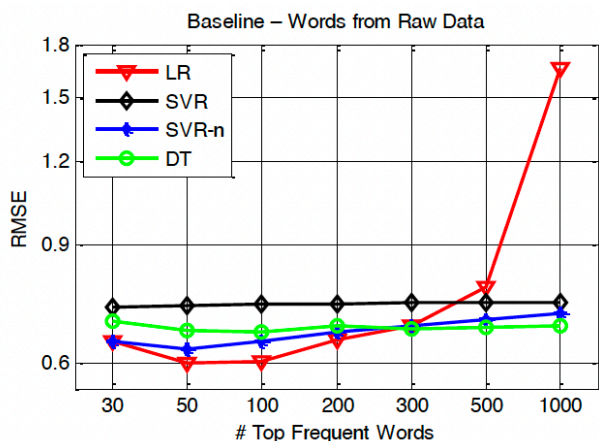


Predicting a Business' Star in Yelp from Its Reviews' Text Alone; Mingming Fan & Maryam Khademi (2014)

• Feature Selection:

- BOW — top K frequency words
- 3 feature vectors:
 - **baseline:** raw data of the top K frequent words in all the reviews
 - **feature engineering I:** POS analysis per sentence, top K words
 - **feature engineering II:** K frequent adjectives
- Predicting star rating as a **regression** task
- Would try to generalize the model in the future by using other business categories (e.g., fashion, beauty) that are not restaurants.

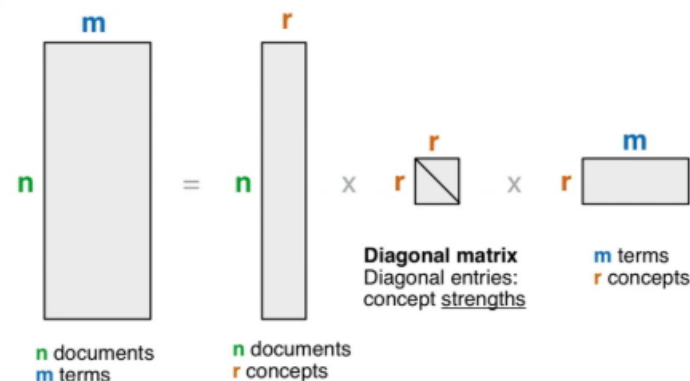
Feature Selection Method	Learning Model	RMSE
Baseline (Top Frequent Words from Raw Data)	Linear Regression	0.6014
	Support Vector Regression	0.7278
	Support Vector Regression-n	0.6296
	Decision Tree	0.6689
Feature Engineering I (Top Frequent Words after POS)	Linear Regression	0.6488
	Support Vector Regression	0.7298
	Support Vector Regression-n	0.6791
	Decision Tree	0.6697
Feature Engineering II (Top Frequent Adjectives after POS)	Linear Regression	0.6052
	Support Vector Regression	0.7135
	Support Vector Regression-n	0.6766
	Decision Tree	0.6766



Yelp Dataset Challenge: Review Rating Prediction;

Nabiha Asghar (2016)

- **Preprocessing:** removed capitalizations, stop words and punctuations
- **Feature Extraction:** feature vector for each review from four different methods:
 - Unigrams (BOW)
 - Unigrams & Bigrams
 - Unigrams, Bigrams, & Trigrams
 - LSI
 - By default, t = the size of vocabulary
 - first t^* topics can be chosen as the most important ones, and thus the top t^* rows of V can be used as the feature matrix
- **ML Models:**
 - Logistic Regression
 - Multinomial Naive Bayes
 - Perceptron ($n_iterations = 50$)
 - SVM (linear, tolerance = 0.001)
- **Results:** Logistic Regression achieved the highest accuracy of 64% using the top 10,000 Unigrams & Bigrams as features, followed very closely by Linear SVC which achieved 63% accuracy using the top 10,000 Unigrams & Bigrams.



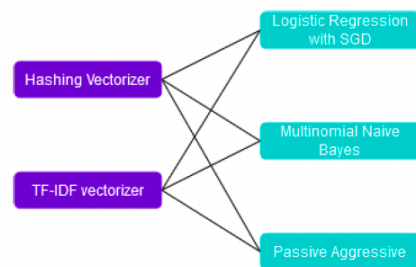
Sentiment Analysis: A Systematic Case Study with Yelp Scores; Wenping Wang et al. (2023)

• Feature Extraction/Engineering:

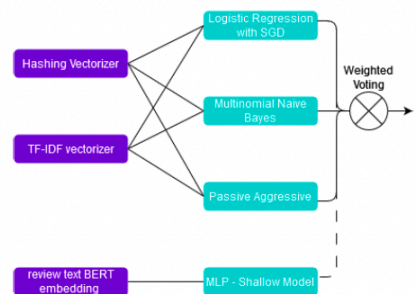
- Hashing Vectorizer (sklearn)
- TF-IDF

• ML Models:

- **Logistic Regression** with SGD I and SGD II
 - **Multinomial Naive Bayes**
 - **Passive Aggressive**
 - Similar to SVM (maximizing the margin of the separating planes between the data points)
 - **(Shallow) Multi-Layer Perceptron (MLP)** with BERT
 - 7 fully connected layers (768, 1024, 512, 512, 256, 64, 32, 5)
 - **(Deep) MLP** with BERT (+ other features)
 - Useful, funny, cool, positive, negative, nchars, nwords, sentiment score
 - Heterogeneous (**ensemble**) classifiers (combined using weighted voting):
 - Baseline models ensemble
 - Baseline models and shallow MLP
 - Baseline models and deep MLP
 - AdaBoost
- For star rating of 5, best model is able to correctly classify 79.57% of the test data points, but it could only correctly classify **32.30% of the data that has ratings of 2**.
- Shallow > deep



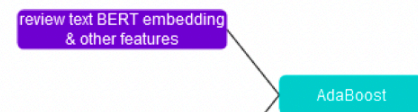
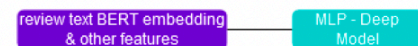
(a) Baseline approaches.



(c) Ensemble approaches. We tried three different combinations of feature extractors and predictors, and use a weighted sum to determine the final prediction.



(b) Multi-layer Perceptron approaches.



(d) AdamBoost approaches.

	Test Accuracy
MLP Shallow Model	0.5307
MLP Deep Model	0.5282
Adaboost I	0.5345
Adaboost II	0.4370
Baseline (hashing) Ensemble	0.5633
Baseline (TF-IDF) Ensemble	0.5689
Baseline (hashing) and MLP Ensemble	0.5832
Baseline (TF-IDF) and MLP Ensemble	0.5837

Review of Findings

- Binary sentiment models (Naive Bayes, Random Forest) good at **positive/negative**
- Multi-class and regression models (Logistic/SVM, Linear Regression) weaker for all 5 star ratings (**accuracy $\leq$ 65%**).
- **TF-IDF and n-gram** features are standard; context-sensitive embeddings (BERT) have emerged for fine-grained results.
- Lexicons help but struggle with **slang, typos, noisy data; sarcasm** and nuanced language largely unsolved.
- **Ensemble methods** sometimes improve results; deep learning gaining ground, especially for multi-aspect analysis and context.
- **Trends observed:**
 - More features $\neq$ better performance (can cause overfitting).
 - Simple > Complex Models

Interesting Findings

- Standard preprocessing and Bag of Words are still strong baselines.
- Too many features can hurt linear regression results; keep it simple when possible.
- Context-aware models (e.g., BERT) for sarcasm, multi-aspect sentiment are promising.
- Aspect-Based Sentiment Analysis worth trying.

Challenges and Gaps

- **Context & Sarcasm:** “Great job burning my pizza!” — an unsolved problem
- **Lexicon mismatch:** limited by real-world data noise, slang, and sarcasm
- *****Imbalanced data***:** most reviews are 4 or 5 stars
- **Explaining middle ratings:** very difficult to discern between 2 & 3 star ratings

Papers Read

Sentiment Analysis on Product Reviews Using Machine Learning Techniques; Rajkumar S. Jagdale, Vishal S. Shirsat and Sachin N. Deshmuk

Yelp Dataset Challenge: Review Rating Prediction; Nabiha Asghar

Sentiment Analysis: A Systematic Case Study with Yelp Scores; Wenping Wang Et al.

Ensemble Sentiment Analysis Using Bi-LSTM and CNN; Puneet Singh Lamba Et al.

Sentiment Analysis of Restaurant Reviews using Combined CNN-LSTM; Naimul Hossain Et al.

Sentiment Analysis of Yelp Reviews by Machine Learning; Hemalatha S, Ramathmika

Sentiment Analysis on Food Review using Machine Learning Approach; Nourin Islam, Ms. Nasrin Akter, Abdus Sattar

Sentiment Analysis using Machine Learning Techniques on Python; Ratheee Et al.

Sentiment Analysis of Yelp's Ratings Based on Text Reviews; Xu Et al.

Predicting a Business' Star in Yelp from Its Reviews' Text Alone; Fan, Khademi